

Problem Set 1

Intermediate Macroeconomics, EC2211

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This problem set is based on material from LN1 and LN2. You are strongly encouraged to work through all questions before Seminar 1, but you should not hand in solutions. The quiz in Lecture 5 will test the same concepts and mechanisms. If you need to prioritize, question 4 on the Solow model is most important.

1 Comparing GDP across countries

(Based on Exercise 7 in Jones, chapter 2) Indian GDP in 2019 was 195 trillion rupees, while U.S. GDP was \$20.6 trillion. The market exchange rate was 70.4 rupees per dollar. India had a lower price level than the United States: after conversion into dollars, the Indian price level was 30.3 percent of the U.S. price level.

- Convert Indian GDP into dollars using the market exchange rate. What was the ratio of Indian GDP to U.S. GDP according to this calculation?
- Calculate the ratio of Indian real GDP to U.S. real GDP after adjusting for the difference in price levels.
- Explain why the answer in part b differs from the answer in part a. Which comparison is more informative about the quantity of goods and services produced in the two countries?

2 Has Finland fallen further behind?

In Lecture Notes 1, we showed that GDP per capita can be written as

$$\frac{Y}{N} = \frac{Y}{H} \times \frac{H}{E} \times \frac{E}{N^{wa}} \times \frac{N^{wa}}{N}.$$

For a variable X , define its annualized percentage growth rate between 2000 and 2025 as

$$g_X = 100 \left[\left(\frac{X_{2025}}{X_{2000}} \right)^{1/25} - 1 \right].$$

Country	Year	Y/N	Y/H	H/E	E/N^{wa}	N^{wa}/N
Finland	2000	76.90	0.10153	1,700.9	0.59090	0.75357
	2025	91.29	0.12274	1,541.5	0.65367	0.73814
Sweden	2000	80.97	0.10057	1,705.3	0.64946	0.72694
	2025	106.40	0.13308	1,621.5	0.68722	0.71747
U.S.	2000	100.00	0.11189	1,711.5	0.71103	0.73438
	2025	138.31	0.16419	1,684.5	0.66743	0.74924

Notes: H/E denotes annual hours per worker. The final two columns are proportions. Working age is defined as ages 15–74.
Source: AMECO and own calculations.

We can then use the approximation

$$g_{Y/N} \approx g_{Y/H} + g_{H/E} + g_{E/N^{wa}} + g_{N^{wa}/N}.$$

The table below reports data for the United States, Sweden, and Finland. The real GDP series has been normalized so that U.S. GDP per capita in 2000 equals 100. The same normalization of real GDP is used when calculating GDP per hour.

- Calculate the annualized growth rates of GDP per capita in Finland, Sweden, and the United States between 2000 and 2025. By how many percentage points per year was growth in Finland slower than in the United States?
- For each country, calculate the contribution from each of the four components to the annualized growth rate of GDP per capita. Which components help account for Finland's slower growth relative to the United States? Which component offsets part of the growth gap?
- Sweden's GDP per capita grew faster than Finland's over the same period. Which components appear most important for explaining the difference?
- In which of the three countries did the change in the working-age share of the population make the largest negative contribution to GDP per capita growth? Explain how this may be related to population aging.

3 What explains growth in output per hour?

The decomposition in the previous question treated output per hour, Y/H , as an observed component of GDP per capita. We now use growth accounting to examine why output per hour changed.

Suppose that production is described by

$$Y_t = A_t K_t^\alpha H_t^{1-\alpha},$$

where H_t is total hours worked. Dividing by hours gives

$$\frac{Y_t}{H_t} = A_t \left(\frac{K_t}{H_t} \right)^\alpha.$$

The table reports capital per hour in Finland, Sweden, and the United States. For each country, capital per hour is expressed as a percentage of capital per hour in the United States in 2000. This normalization does not affect the calculated growth rates.

Country	Capital per hour, K/H	
	2000	2025
Finland	110.4	174.0
Sweden	106.4	163.7
United States	100.0	156.9

Source: AMECO, Penn World Table 11.0, and own calculations. The latest available PWT observations, from 2023, are used as proxies for 2025.

Assume that the capital share is $\alpha = 1/3$.

- Calculate $g_{K/H}$, the annualized growth rate of capital per hour, in Finland, Sweden, and the United States between 2000 and 2025. Compare the rates of capital deepening across the three countries.
- Growth accounting gives the approximation

$$g_{Y/H} \approx g_A + \alpha g_{K/H}.$$

Using your estimates of $g_{Y/H}$ from the previous question, decompose productivity growth in each country into a contribution from capital deepening and a contribution from TFP growth.

- Does this calculation show that differences in technological progress caused the differences in productivity growth? Explain what else may be captured by measured TFP growth.

4 Two changes in the labor force in the Solow model

Consider the Solow model with

$$Y_t = AK_t^\alpha L_t^{1-\alpha}, \quad K_{t+1} = (1 - \delta)K_t + sY_t.$$

The economy initially is in a steady state. Throughout the question, A , s , δ , and α are constant.

A one-time increase in the level of the labor force

At date $t = 0$, the labor force rises immediately and permanently from L to $L' > L$. After this one-time change, the labor force remains constant at L' .

- What happens immediately to aggregate capital K_0 , capital per worker k_0 , aggregate output Y_0 , and output per worker y_0 ? Distinguish carefully between quantities that are pre-determined and quantities that can change immediately.

- b. Draw the Solow diagram in per-worker terms. Does either the investment curve sAk^α or the depreciation line δk shift when the labor force increases?
- c. Describe the transition after the increase in the labor force. What happens to the growth rate of output per worker during the transition?
- d. Compare the initial and final steady states. What happens in the long run to capital and output per worker? What happens to aggregate capital and aggregate output?

A higher population growth rate

Now return to the initial steady state and instead suppose that the growth rate of the labor force rises permanently from n to $n' > n$.

- e. Draw the Solow diagram with population growth. Which curve shifts when n rises?
- f. What happens during the transition to capital and output per worker? What are the long-run effects on their levels and growth rates?
- g. What is the long-run growth rate of aggregate output before and after the change?
- h. Summarize the key difference between a one-time increase in the *level* of the labor force and a permanent increase in its *growth rate*.

Check your understanding

For each statement, decide whether it is true or false and explain your answer in one or two sentences.

1. If a country has a lower price level than the United States, converting its GDP at the market exchange rate will tend to understate its real GDP relative to the United States.
2. In growth accounting, the contribution from capital deepening is $\alpha g_{K/H}$ rather than $g_{K/H}$.
3. The TFP residual is a direct measure of technological progress and cannot reflect measurement error or omitted inputs.
4. A one-time permanent increase in the level of the labor force lowers steady-state output per worker in the basic Solow model.
5. A higher population growth rate lowers steady-state output per worker but raises the long-run growth rate of aggregate output when TFP is constant.

Recommended extra exercises

Exercises marked by a * are solved in the textbook.

- Jones chapter 2
 - Exercise *2: What accounts as GDP (I)
 - Exercise 3: What accounts as GDP (II)
 - Exercise 9: Earthquakes and GDP

Suppose the rural part of a country is hit by a major earthquake that destroys 10 percent of the country's housing stock. The government and private sector respond with a major construction effort to help rebuild houses. Discuss how this episode is likely to affect (a) the economic well-being of the people in the country and (b) the economy's measured GDP.
- Jones chapter 3
 - Exercise *6: U.S. growth
 - Exercise 9: Computing growth rates (I)
 - Exercise 10: Computing growth rates (II)
- Jones chapter 4
 - Exercise *1: Returns to scale in production
 - Exercise 2: The "per person" versions of production functions
 - Exercise *3: The Black Death
- Jones chapter 5
 - Exercise 1: A decrease in the investment rate

Suppose a country enacts a tax policy that discourages investment, and the policy reduces the investment rate immediately and permanently from s to s' . Assuming the economy starts in its initial steady state, use the Solow model to explain what happens to the economy over time and in the long run. Draw a graph showing how output evolves over time (put y_t on the vertical axis and time on the horizontal axis), and explain what happens to economic growth over time.
 - Exercise *3: An earthquake

(Use the Solow model to analyze the consequences of an earthquake that destroys part of the capital stock...)